

TEAM CONTRACT – TEAM 2

Project Title: Autonomous Smart Car with Vision-Guided Navigation & Mapping

Team Name: Vision & Control Systems Group

Team Members:

1) Joseph Toma

2) Evan Romano

3) Mohammed Alamen Qassab

4) _____

5) _____

Project Vision

As a team, our goal is to design and build an autonomous smart car that can navigate an obstacle-filled environment using onboard vision and decision-making. The project focuses on combining mechanical design, electronics, and software to create a system that can perceive its surroundings, build an internal understanding of the track, and select efficient paths toward a target.

In addition to autonomy, we aim to improve the car's reliability by addressing known mechanical, wiring, and power-related issues identified in previous smart car designs. Through this project, we will apply practical principles of mechatronics and AI engineering, including system integration, robustness, and safety. The final system is intended to be functional, well-documented, and suitable for real-world testing and future development.

Team Procedures

1. Day, time, and place for regular **team meetings**:

The team will hold a primary weekly meeting every Monday from 12:30 PM to 2:30 PM. This meeting will be used to evaluate progress, discuss technical challenges, and prepare for the upcoming lab. In addition, shorter meetings will be held throughout the week on Discord, scheduled based on mutual availability, to address specific tasks or issues. All members will also attend the weekly lab session on Mondays from 6:30 PM to 9:30 PM.

2. Preferred method of **communication** (e.g., e-mail, cell phone, wired phone, Blackboard Discussion Board, face-to-face, in a certain class) in order to inform each other of team meetings, announcements, updates, reminders, and problems:

Discord will be the primary method of communication for discussions, updates, and meeting coordination. Official files and documents will be stored on the created course Microsoft Teams page in the designated team folder. A GitHub repository will be created to manage all code and CAD models developed during the project. In-person communication will occur during scheduled meetings and lab sessions.

3. **Decision-making policy** (by consensus? by majority vote?):

When disagreements arise, members will pitch their ideas to the team. If consensus cannot be reached, a voting system will be used. If neutral members are present, they will vote on the proposed options. If no neutral members are available, decisions will be made by majority vote. Majority rule will apply in all final decisions.

4. Method for setting and following meeting **agendas** (Who will set each agenda? When? How will team members be notified/reminded? Who will be responsible for the team following the agenda during a team meeting? What will be done to keep the team on track during a meeting?):

Mohammed Alamen Qassab will be responsible for setting the weekly meeting agenda. Agendas and important communications for the upcoming week will be shared by the weekend prior to the meeting. During meetings, Mohammed will ensure the team stays on track with the agenda while still allowing flexibility for members to lead discussions related to their assigned responsibilities or areas of focus.

5. Method of **record keeping** (Who will be responsible for recording & disseminating minutes? How & when will the minutes be disseminated? Where will all agendas & minutes be kept?):

Evan Kale Romano will be responsible for recording meeting minutes using a template from previous project-based courses. All meeting minutes will be stored in a dedicated "Meeting Minutes" subfolder on the team's Microsoft Teams page. The team will also maintain a shared task tracker using Google Sheets, which will include task descriptions, deadlines, urgency levels, and assigned members to ensure transparency and accountability.

Team Expectations

Work Quality

1. **Project standards** (What is a realistic level of quality for team presentations, collaborative writing, individual research, preparation of drafts, peer reviews, etc.):

All work will meet undergraduate engineering standards and align with the project rubric, with particular focus on achieving Level 4 performance criteria. Deliverables are expected to demonstrate strong effort, technical accuracy, and clear documentation, while accounting for time availability and overall course load.

2. **Strategies** to fulfill these standards:

All work completed by individual members will be thoroughly reviewed by the rest of the team. Feedback from teaching assistants and, when possible, the instructor will be actively sought prior to submission. Deliverables will be evaluated against the rubric to ensure all requirements are met or exceeded. This process will be combined with regular testing, documentation, and iteration to maintain quality throughout the project.

Team Participation

1. Strategies to ensure cooperation and equal distribution of tasks:

The shared task tracker will be used to monitor workload distribution and ensure tasks are assigned fairly. If a member completes their work early and has the capacity, they may be reassigned to assist teammates with heavier workloads or those encountering difficulties.

2. Strategies for encouraging/including ideas from all team members (team maintenance):

The team will operate democratically. Members with experience in specific areas may take the lead on related tasks, but all team members are encouraged to contribute ideas during open discussions. Any member may pitch an idea regardless of expertise, provided it is clearly explained. Decisions will follow the agreed-upon voting process when necessary.

3. Strategies for keeping on task (task maintenance):

Progress will be tracked using the task tracker. In addition, Mohammed will regularly check in with team members throughout the week outside of scheduled meetings. These check-ins may include informal Discord meetings or brief status updates to ensure progress is being made and issues are addressed early.

4. Preferences for leadership (informal, formal, individual, shared):

Leadership will be shared among the team. Specific leads may be assigned for particular subsystems or tasks based on expertise, but responsibility and decision-making remain collaborative.

Personal Accountability

1. Expected individual attendance, punctuality, and participation at all team meetings:

All members are expected to attend meetings on time and participate actively. If a member cannot attend, they must notify the team at least 24 hours in advance when possible. In the case of emergencies, notification may occur afterward.

2. Expected level of responsibility for fulfilling team assignments, timelines, and deadlines:

Each member is responsible for completing their assigned tasks listed in the task tracker on time. If issues or delays arise, the member must communicate them early so the team can adjust responsibilities or timelines accordingly.

3. Expected level of communication with other team members:

Team members are expected to communicate clearly and respectfully using the agreed-upon platforms. Messages should be responded to within a reasonable time frame, typically within 24 hours.

4. Expected level of commitment to team decisions and tasks.

Full commitment to team decisions and assigned tasks is expected. While delays may occur due to heavy engineering course loads, this does not excuse neglect of responsibilities. Repeated failures to contribute will be addressed by the team.

Consequences for Failing to Follow Procedures and Fulfill Expectations

1. Describe, as a group, how you would handle **infractions** of any of the obligations of this team contract:

Initial issues will be addressed through calm and respectful group discussion to clarify expectations and resolve problems. The goal will be to support the member and restore productive collaboration.

2. Describe what your team will do **if the infractions continue**:

If issues persist, they will be documented and may be brought to the attention of the instructor or teaching assistant, as appropriate.

- a) *I participated in formulating the standards, roles, and procedures as stated in this contract.*
b) *I understand that I am obligated to abide by these terms and conditions.*
c) *I understand that if I do not abide by these terms and conditions, I will suffer the consequences as stated in this contract.*

- 1) Joseph Toma Date January 16th, 2026
2) Evan Romano Date January 16th, 2026
3) Mohammed Alamen Qassab Date January 16th, 2026
4) _____ Date _____
5) _____ Date _____